

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 753 Computer Vision — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Szeliski — Computer Vision: Algorithms and Applications, 2nd edition (free PDF)	The primary source, free in full from the author. The geometric chapters are the reference for Modules 02 through 07 and the second edition covers the learned material of Modules 08–10.
Hartley & Zisserman — Multiple View Geometry	The definitive treatment of Modules 04 and 05. Dense and worth the effort; the sample chapters are free and the errata page is useful. Library copy for the rest.
Michigan EECS 498/598 — Deep Learning for Computer Vision (free lectures and assignments)	The best free course for Modules 08 through 10. Full video lectures, slides, and assignments, taught at the right level for someone who has already done CSCE 636.
Mildenhall et al. — NeRF; and Kerbl et al. — 3D Gaussian Splatting (both free)	The two papers Module 11 is built on. Both are readable and both have reference implementations, and the contrast between them is instructive.
OpenCV documentation and tutorials (free)	The working toolkit for Modules 02–07. The calibration and findEssentialMat tutorials are worth following exactly once before writing your own.
COLMAP documentation (free, open source)	A production structure-from-motion pipeline you can read. Module 05's reference point, and what your own implementation should be compared against.

Module-by-module map

Module 01 — The Inverse Problem

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 1 (free PDF)	The framing of §1 and a survey of the whole subject, with a good history section.
Belhumeur, Kriegman & Yuille — The Bas-Relief Ambiguity (free)	The proved ambiguity from §1's table , and worth reading as an example of establishing that something is impossible.
Marr — Vision (1982), chapter 1	Where the inverse-problem framing of this module originates . Dated in its specifics and still the clearest statement of the problem. Library copy.
Hartley & Zisserman — Multiple View Geometry, chapter 1 (free sample)	The projective-geometry preview, which Module 02 develops properly.

Module 02 — Cameras and Calibration

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 2 (free PDF)	Image formation, camera models, and distortion — the reference for §1 and §2.
Zhang — A Flexible New Technique for Camera Calibration (free)	The method §3 describes , and what OpenCV implements. Short and worth reading in full.
OpenCV — camera calibration tutorial (free)	The working procedure, with code. Follow it once exactly before deviating.
Hartley & Zisserman — Multiple View Geometry, chapters 2–6	Projective geometry and camera models done properly , including the duality of §1's callout.

Module 03 — Features and Correspondence

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 7 (free PDF)	Feature detection, description, and matching — the reference for §1 through §3.
Lowe — Distinctive Image Features from Scale-Invariant Keypoints (free)	SIFT, in the author's own words . The ratio test of §3 is introduced here, with the distance distributions that justify it.
Fischler & Bolles — Random Sample Consensus (free)	The original RANSAC paper , including the iteration-count derivation of §3.
Sarlin et al. — SuperGlue (free)	The §4 result . The attention-plus-assignment architecture, and the wide-baseline evaluation.

Module 04 — Two-View Geometry

Resource	What it covers for this module
Hartley & Zisserman — Multiple View Geometry, chapters 9–12	The definitive treatment of this module. Epipolar geometry, F and E, triangulation, and the degeneracies of §4.
Nistér — An Efficient Solution to the Five-Point Relative Pose Problem (free)	The 5-point solver of §1's table , and why the sample size matters so much.
Szeliski — Computer Vision, chapter 11 (free PDF)	A gentler path through the same material, with the practical pipeline.
Hartley — In Defense of the Eight-Point Algorithm (free)	Why normalisation of coordinates matters so much — CSCE 620's and CSCE 669's conditioning lesson, in this exact setting.

Module 05 — Structure from Motion

Resource	What it covers for this module
Schönberger & Frahm — Structure-from-Motion Revisited (COLMAP) (free)	The reference pipeline of §1 , with the seed selection and verification details that matter most.
Triggs et al. — Bundle Adjustment: A Modern Synthesis (free)	The §2 reference. Long, thorough, and the source for the sparsity and gauge-fixing treatment.
Ceres Solver documentation — bundle adjustment example (free)	A working sparse bundle adjustment you can read and run , with the Schur complement option exposed.
Cadena et al. — Past, Present, and Future of SLAM (free survey)	The §3 material in its SLAM form , including loop closure and place recognition.

Module 06 — Stereo and Dense Depth

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 12 (free PDF)	Stereo correspondence, rectification, and the cost volume — the reference for §1 and §2.
Hirschmüller — Stereo Processing by Semi-Global Matching (free)	SGM, the method of §2's table. Implementable directly from the paper.
Scharstein & Szeliski — the Middlebury stereo benchmark (free)	Ground-truth data and a leaderboard , plus the taxonomy that organises §2's methods.
Zabih & Woodfill — Non-parametric Local Transforms (census) (free)	The census transform of §1 , and the monotonic-invariance argument.

Module 07 — Motion and Tracking

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 9 (free PDF)	Motion estimation, optical flow, and the pyramid scheme — the reference for §1 and §2.
Baker & Matthews — Lucas-Kanade 20 Years On (free)	The §1 method, developed properly , including the inverse-compositional formulation that makes it fast.
Welch & Bishop — An Introduction to the Kalman Filter (free)	The clearest free introduction to §3 , with the assumptions stated explicitly.
Teed & Deng — RAFT: Recurrent All-Pairs Field Transforms (free)	The learned flow method that addresses §2's intrinsic failure , by correlating all pairs rather than searching locally.

Module 08 — Learned Representations

Resource	What it covers for this module
Michigan EECS 498 — CNN architectures and transfer lectures (free)	The reference for this module. The transfer-learning lecture covers §2 directly.
Zeiler & Fergus — Visualizing and Understanding Convolutional Networks (free)	Where §1's hierarchy picture comes from , and worth reading alongside §4's caveat about it.
Araujo, Norris & Sim — Computing Receptive Fields of Convolutional Neural Networks (free, Distill)	The §3 recurrence, worked out , with interactive calculators.
Touvron et al. — Fixing the train-test resolution discrepancy (free)	The §3 bug, measured and fixed — a good example of a data-pipeline problem mistaken for a model problem.

Module 09 — Detection and Segmentation

Resource	What it covers for this module
Michigan EECS 498 — detection and segmentation lectures (free)	The reference for §1 and §4 , with the architecture families laid out clearly.
Lin et al. — Focal Loss for Dense Object Detection (free)	The §1 imbalance problem and its fix , with the analysis that motivates it.
Carion et al. — End-to-End Object Detection with Transformers (DETR) (free)	The matching loss of §1 — read it for the loss, which is the contribution.
COCO detection evaluation documentation (free)	Exactly what the metrics of §3 compute , including the size breakdown that is so often ignored.

Module 10 — Transformers and Self-Supervision

Resource	What it covers for this module
Dosovitskiy et al. — An Image is Worth 16x16 Words (ViT) (free)	The §1 architecture , including the data-volume crossover measurements of the comparison table.
Caron et al. — Emerging Properties in Self-Supervised Vision Transformers (DINO) (free)	The §2 self-distillation method , and the attention maps that segment objects unsupervised.
He et al. — Masked Autoencoders Are Scalable Vision Learners (free)	The §2 masked objective , with the probe-versus-fine-tune asymmetry reported honestly.
Radford et al. — Learning Transferable Visual Models from Natural Language Supervision (CLIP) (free)	The §3 result , and the limitations section is unusually candid and worth reading.

Module 11 — Neural Scene Representations

Resource	What it covers for this module
Mildenhall et al. — NeRF: Representing Scenes as Neural Radiance Fields (free)	The §1 and §2 method . Readable, with a reference implementation and the positional-encoding ablation that makes §2's point.
Kerbl et al. — 3D Gaussian Splatting for Real-Time Radiance Field Rendering (free)	The §3 method , with the adaptive density control described properly.
Müller et al. — Instant Neural Graphics Primitives (free)	The multiresolution hash encoding that made radiance fields train in seconds — and a good example of a data-structure contribution beating an architectural one.
Tewari et al. — Advances in Neural Rendering (free survey)	The field laid out systematically , including the surface-versus-volume question of §2.

Module 12 — Evaluation and Failure

Resource	What it covers for this module
Recht et al. — Do ImageNet Classifiers Generalize to ImageNet? (free)	The §1 replication . The measured gap, with the collection protocol described in enough detail to assess.
Hendrycks & Dietterich — Benchmarking Neural Network Robustness to Common Corruptions (free)	The corruption benchmark of §4 , and the finding that clean accuracy does not predict corrupted accuracy.
Geirhos et al. — Shortcut Learning in Deep Neural Networks (free)	The §2 reference , with the documented cases and the detection strategies.
Madry et al. — Towards Deep Learning Models Resistant to Adversarial Attacks (free)	The §3 phenomenon treated rigorously , with the threat model stated explicitly — which is the part worth imitating.

Module 13 — Vision in Systems

Resource	What it covers for this module
Li, Wang & Ramanan — Towards Streaming Perception (free)	The §1 metric. Evaluates perception against the world's current state rather than the frame's, which is the right formulation.
NVIDIA TensorRT and ONNX Runtime documentation (free)	The practical tools for §1's inference row , including quantisation and the profiling output that shows where the time goes.
Sculley et al. — Hidden Technical Debt in Machine Learning Systems (free)	The §3 argument , and the clearest short statement of why monitoring is not optional.
Szeliski — Computer Vision, chapter 14 and the conclusions (free PDF)	A survey of applications and open problems , and a reasonable place to choose what to study next.